

Equal-Well Allen–Cahn Fronts: Translation Uniqueness and a Pöschl–Teller Spectral Atlas

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29 August 2026 (revision 0)

Abstract

We close a source-local theorem for the one-dimensional equal-well Allen–Cahn reaction–diffusion equation. For $u_t = u_{xx} + \varepsilon^{-2}(u - u^3)$, every monotone heteroclinic from -1 to $+1$ is a translate of $\tanh(\xi/(\sqrt{2}\varepsilon))$, and the equal-well balance forces its travelling speed to be zero. The gradient-flow energy dissipates exactly, with interfacial energy $2\sqrt{2}/(3\varepsilon)$. Linearization is the factorized Pöschl–Teller operator: translation is the simple zero mode, the shape mode has eigenvalue $-3/(2\varepsilon^2)$, and the essential spectrum is $(-\infty, -2/\varepsilon^2]$. Singular scaling faces are stated explicitly. The front is heteroclinic, not a primitive periodic orbit, so this is a rigorous Route-A negative result rather than an arithmetic or Hilbert–Pólya claim.

1 Frozen equation and the heteroclinic

We use the scaled family

$$u_t = u_{xx} + \varepsilon^{-2}(u - u^3), \quad \varepsilon > 0, \quad (1)$$

whose $\varepsilon = 1$ member is the requested Allen–Cahn equation. Put $W(q) = (1 - q^2)^2/4$; then (1) is the L^2 gradient flow of

$$E_\varepsilon[u] = \int_{\mathbb{R}} \left(\frac{1}{2} u_x^2 + \varepsilon^{-2} W(u) \right) dx. \quad (2)$$

For a travelling ansatz $u(x, t) = U(\xi)$, $\xi = x - ct$, the profile equation is

$$U'' + cU' + \varepsilon^{-2}(U - U^3) = 0, \quad U(-\infty) = -1, \quad U(+\infty) = 1. \quad (3)$$

Theorem 1 (front, speed selection, and translation uniqueness). *Every nonconstant monotone C^2 solution of (3) has $c = 0$ and, for some $\xi_0 \in \mathbb{R}$,*

$$U(\xi) = U_\varepsilon(\xi - \xi_0) := \operatorname{anh}\left(\frac{\xi - \xi_0}{\sqrt{2}\varepsilon}\right). \quad (4)$$

The reverse orientation is $-U_\varepsilon$. For the increasing orientation,

$$\frac{1}{2}(U'_\varepsilon)^2 = \varepsilon^{-2}W(U_\varepsilon), \quad \int_{\mathbb{R}} (U'_\varepsilon)^2 d\xi = E_\varepsilon[U_\varepsilon] = \frac{2\sqrt{2}}{3\varepsilon}. \quad (5)$$

Proof. Multiply (3) by U' and integrate from $-\infty$ to $+\infty$. The endpoint derivatives and $W(\pm 1)$ vanish, giving

$$c \int_{\mathbb{R}} (U')^2 d\xi = 0.$$

Since a nonconstant front has positive integral, $c = 0$. A first integral is then $(U')^2/2 = \varepsilon^{-2}W(U)$. Monotonicity selects the positive square root, and separation of variables gives (4) with one free translation. The identity $U'_\varepsilon = (\sqrt{2}\varepsilon)^{-1} \operatorname{sech}^2 y$, $y = (\xi - \xi_0)/(\sqrt{2}\varepsilon)$, and $\int_{\mathbb{R}} \operatorname{sech}^4 y dy = 4/3$ prove (5). $\square$

Source note

The phase-field context is recorded by Allen and Cahn [1]; the travelling-front convergence context is recorded by Fife and McLeod [2]. The formulas, spectrum, and boundaries above are re-derived in this artifact; the references are contextual and do not enlarge the claim boundary.

References

- [1] S. M. Allen and J. W. Cahn, A microscopic theory for antiphase boundary motion and its application to antiphase domain coarsening, *Acta Metallurgica* 27(6), 1085–1095 (1979). DOI: 10.1016/0001-6160(79)90196-2.
- [2] P. C. Fife and J. B. McLeod, The approach of solutions of nonlinear diffusion equations to travelling front solutions, *Archive for Rational Mechanics and Analysis* 65(4), 335–361 (1977). DOI: 10.1007/BF00250432.

Declarations

Scope. NO_BAD_EULER_OR_ROOT_NUMBER; no target arithmetic, periodic-orbit, or operator claim. **Data and code.** The exact front, energy ledger, independent checks, and build instructions are released with HCS-C231. **AI-use disclosure.** Generative tools assisted drafting and code generation; the internal artifact chain checked displayed claims and metadata. This is not external peer review.